DiffCR: A Fast Conditional Diffusion Framework for Cloud Removal From Optical Satellite Images

Xuechao Zou, *Graduate Student Member, IEEE*, Kai Li, *Student Member, IEEE*, Junliang Xing, *Senior Member, IEEE*, Yu Zhang, Shiying Wang, Lei Jin, and Pin Tao, *Member, IEEE*

***Abstract*— Optical satellite images are a critical data source; however, cloud cover often compromises their quality, hindering image applications and analysis. Consequently, effectively removing clouds from optical satellite images has emerged as a prominent research direction. Recent advances in deep learning-based cloud removal methods have been significant, but the quality of image generation still needs to be improved. Diffusion models have demonstrated remarkable success in diverse image generation tasks, showcasing their potential in addressing this challenge. This article presents a novel framework called DiffCR, which leverages conditional guided diffusion with deep convolutional networks for high-performance cloud removal for optical satellite imagery. Specifically, we introduce a decoupled encoder for conditional image feature extraction, providing a robust color representation to ensure the close similarity of appearance information between the conditional input and the synthesized output. Moreover, we propose a novel and efficient time and condition fusion block (TCFBlock) within the cloud removal model to accurately simulate the correspondence between the appearance in the conditional image and the target image at a low computational cost. Extensive experimental evaluations on three commonly used benchmark datasets demonstrate that DiffCR consistently achieves the state-of-the-art (SOTA) performance on all metrics, with parameter and computational complexities amounting to only 5.1% and 5.4%, respectively, of those previous best methods. The source code, pretrained models, and all the experimental results will be publicly available at https://github.com/XavierJiezou/DiffCR upon the paper's acceptance of this work.**

***Index Terms*— Autoencoder, cloud removal, deep learning, diffusion models, optical satellite images.**

I. Introduction

IN RECENT years, with the rapid advancement of remote sensing technology, satellite imagery, particularly optical satellite images, has found extensive applications across various domains, including agriculture monitoring [1], ground target detection [2], [3], and land cover classification [4], [5]. However, cloud interference during the imaging process poses a significant challenge. According to the International Satellite Cloud Climatology Project (ISCCP) data, it was found that an average of 67% of the Earth's surface is covered by clouds [6]. The opacity of thick clouds obstructs the underlying object information, drastically reducing their utility and effectiveness, and severely hampers further satellite imagery analysis and application. Therefore, developing an effective method to remove clouds from optical remote sensing images has become an essential and pressing research problem, as a successful cloud removal algorithm would significantly enhance data availability and improve the accuracy of downstream tasks dependent on optical satellite images.

With the advancement of deep learning techniques, generative adversarial networks (GANs) have emerged as a prominent research direction in the field of image processing. GAN-based methods can learn data distributions and generate high-quality samples, finding successful applications in tasks such as image super-resolution [7], [8], [9], image inpainting [10], and style transfer [11]. In the context of cloud removal, GAN-based methods [12], [13], [14], [15], [16] have also been widely employed, yielding promising results. Specifically, they consist of two key components: a generator responsible for generating cloud-free surface images from cloud-contaminated images and a discriminator tasked with determining the authenticity of the generated images. The generator and discriminator gradually improve their performance through iterative training, generating more realistic and clear surface images.

Despite the remarkable performance of GAN-based and other deep learning-based approaches in cloud removal, several challenges remain. For instance, they are inherently difficult to train and can suffer from issues such as mode collapse, leading to distorted or repetitive generated images. Moreover, cloud morphology and distribution exhibit diversity and complexity, placing complex demands on GAN-based methods training and application. Therefore, designing more robust and efficient deep generative models remains crucial in cloud removal research.

To achieve cloud removal while preserving high image fidelity, we propose to explore the applicability of diffusion models for cloud removal in optical satellite images. The diffusion-based approach constructs data labels by a

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Xuechao Zou, Yu Zhang, and Shiying Wang are with the Department of Computer Technology and Applications, Qinghai University, Xining 810016, China (e-mail: xuechaozou@foxmail.com; tinkzy@163.com; wangshiying.qhu@foxmail.com).

Kai Li and Junliang Xing are with the Department of Computer Science and Technology, Tsinghua University, Beijing 100190, China (e-mail: tsinghua.kaili@gmail.com; jlxing@tsinghua.edu.cn).

Lei Jin is with the School of Electronic Engineering, Beijing University of Posts and Telecommunications, Beijing 100876, China (e-mail: jinlei@bupt.edu.cn).

Pin Tao is with the Department of Computer Science and Technology and the Key Laboratory of Pervasive Computing, Ministry of Education, Tsinghua University, Beijing 100190, China (e-mail: taopin@tsinghua.edu.cn).

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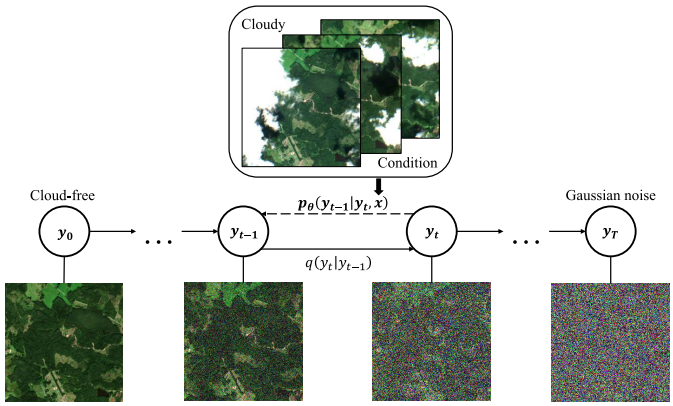


Fig. 1. Overall pipeline of our proposed cloud removal framework DiffCR. Training: this phase consists of the forward process (from left to right) and the reverse process (from right to left). The forward process, also known as the diffusion process, progressively perturbs the ground truth $\mathbf{y}_0$ by adding Gaussian noise to control the training of the parameterized neural network p_θ through an intermediate transition distribution. The reverse process, also known as the denoising process, takes the noisy image $\mathbf{y}_t$ as input to p_θ and iteratively refines $\hat{\mathbf{y}}_0$. In addition, cloudy images $\mathbf{x}$ and noise level t are conditionally injected into p_θ to guide the generation of cloud-free images. Inference: this phase takes the Gaussian noise $\mathbf{y}_T$ and cloudy images $\mathbf{x}$ and t as input to pretrained p_θ to iteratively reconstruct the cloud-free image $\hat{\mathbf{y}}_0$.

diffusion process with added noise. An iterative refinement process follows this through inverse denoising to produce finer samples. In this manner, rather than modeling complex cloud-free features in one go, our proposed cloud removal framework DiffCR (see Fig. 1) decomposes the problem into a series of forward–backward diffusion steps to learn reasonable cloud-free details. With the aforementioned optimizations and innovations, DiffCR achieves the generation of high-quality samples in a mere single step and complete convergence in 3–5 steps. Our main contribution is to propose a decoupled conditional diffusion architecture consisting of TFCBlock units, and details are given as follows.

1) We present the DiffCR, a novel and fast conditional diffusion framework for cloud removal from optical satellite images. This framework decouples the processing of conditional information, enabling the generation of high-fidelity images in just one sampling step.
2) We propose an efficient and novel time and condition fusion block (TCFBlock) as the basic unit of the conditional denoising model that can explicitly integrate the multiscale features of noisy images, multiscale condition embedding, and implicit time embedding.

We demonstrate the empirical validity of DiffCR by comparing it with various competitive GAN, regression, and diffusion-based methods on three commonly used benchmark datasets (Sen2_MTC_New [15], WHUS2-CRv [17], and SEN12MS-CR [18]) showing state-of-the-art (SOTA) results. Besides, we conduct extensive ablation experiments (see Table I) to investigate the impact of various architectural designs in denoising autoencoder. We believe that our work will contribute to efficient real-time cloud removal research, achieving high fidelity and robustness, even with limited computational resources, and providing valuable insights for applying conditional diffusion models in other computer vision domains and generative artificial intelligence, such as text-to-image and image-to-image synthesis.

II. Related Work

A. Cloud Removal

Based on the input data, cloud removal methods can be categorized into single-temporal and multitemporal approaches. Due to data limitations (lacking a globally distributed multitemporal cloud removal dataset), early cloud removal methods [12], [19], [20], [21], [22], [23] based on optical satellite imagery primarily focused on single-temporal approaches, aiming to remove clouds and restore land features from a single cloud-contaminated image at a specific geographic location. However, obtaining a cloud-free image from a single frame becomes a significant challenge when the cloud coverage is extensive, as there is insufficient information to facilitate image reconstruction.

With advancements in remote sensing technology, satellites can capture images of the same location at shorter intervals. This advancement lets us easily acquire multiple satellite images captured simultaneously, providing ample data support for designing effective cloud removal methods. Generally, cloud positions vary over time, and the regions obscured by clouds at the same geographic location do not entirely overlap at different moments. Leveraging spectral and temporal information, existing multitemporal cloud removal methods [14], [15], [24], [25], [26] fuse multiple cloudy images to generate detail-rich cloud-free images.

Existing cloud removal approaches have predominantly relied on GANs, enhancing image reconstruction by introducing adversarial losses during training. MCGAN [12] extended CGANs from RGB images to multispectral images for cloud removal. Spa-GAN [21] introduced spatial attention mechanisms in GANs to improve information recovery in cloud regions, resulting in higher quality cloud-free images. AE [24] employed convolutional autoencoders trained on multitemporal remote sensing datasets for cloud removal. STNet [25] integrated cloud detection techniques and fused spatiotemporal features from multiple cloudy images for cloud removal. STGAN [14] regarded cloud removal as a conditional image synthesis problem and proposed a spatiotemporal generative network. CTGAN [15] introduced a Transformer-based GAN for cloud removal. PMAA [26] achieved efficient cloud removal using progressive autoencoders. In addition, there are also some works [16], [27], [28] that utilize radar imagery to assist in cloud removal, as well as a limited number of works [29], [30] that employ diffusion models. However, due to the limitations of GAN frameworks and adversarial learning mechanisms, existing GAN-based methods struggle to attain satisfactory image fidelity. Moreover, although a few diffusion-based methods have performed well, they are often highly inefficient, typically requiring thousands of sampling steps. Therefore, this work explores a new and fast cloud removal framework using conditional diffusion models, achieving superior fidelity without bells and whistles.

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TABLE I
QUANTITATIVE COMPARISON OF THE CLOUD REMOVAL PERFORMANCE AND EFFICIENCY OF ARCHITECTURE CHANGES IN DENOISING AUTOENCODER

Width	Block	Depth			Skip Connection	PSNR ↑	SSIM ↑	FID ↓	LPIPS ↓	Params (M) ↓	MACs (G) ↓
		Encoder	Middle	Decoder							
16	TCFBlock [Ours]	[1,1,1,1]	1	[1,1,1,1]	Sum	17.789	0.620	104.107	0.359	1.67	3.11
32	TCFBlock [Ours]	[1,1,1,1]	1	[1,1,1,1]	Sum	18.706	0.654	85.439	0.305	6.01	11.79
64	TCFBlock [Ours]	[1,1,1,1]	1	[1,1,1,1]	Sum	19.150	0.671	83.162	0.291	22.91	45.86
96	TCFBlock [Ours]	[1,1,1,1]	1	[1,1,1,1]	Sum	19.194	0.657	84.796	0.283	50.80	102.21
128	TCFBlock [Ours]	[1,1,1,1]	1	[1,1,1,1]	Sum	19.338	0.652	86.075	0.278	89.67	180.85
64	ConvNeXt [61]	[1,1,1,1]	1	[1,1,1,1]	Sum	18.042	0.650	113.985	0.325	25.61	51.18
64	ResNet [53]	[1,1,1,1]	1	[1,1,1,1]	Sum	18.966	0.664	90.542	0.300	42.90	74.56
64	NAFBlock [60]	[1,1,1,1]	1	[1,1,1,1]	Sum	19.042	0.671	83.689	0.295	23.78	45.87
64	TCFBlock [Ours]	[1,1,1,1]	2	[1,1,1,1]	Sum	19.205	0.670	86.717	0.292	30.02	47.48
64	TCFBlock [Ours]	[1,1,1,1]	3	[1,1,1,1]	Sum	19.321	0.671	89.161	0.294	37.13	49.09
64	TCFBlock [Ours]	[1,1,3,1]	1	[1,1,1,1]	Sum	19.285	0.658	101.486	0.290	24.78	58.90
64	TCFBlock [Ours]	[1,1,3,1]	1	[1,3,1,1]	Sum	19.274	0.677	74.908	0.273	25.78	62.16
64	TCFBlock [Ours]	[1,1,1,1]	1	[1,1,1,1]	Concat [73]	15.134	0.337	236.843	0.507	23.61	48.03
64	TCFBlock [Ours]	[1,1,1,1]	1	[1,1,1,1]	Attention [83]	18.017	0.619	121.777	0.354	25.97	55.36

B. Diffusion Models

Image generation has been a popular research direction in computer vision, with various image generation paradigms [31], [32], [33] proposed and applied to this task. In recent years, numerous GAN-based models have dominated the SOTA. However, this paradigm has been disrupted by the advent of diffusion models.

Inspired by thermodynamics [34], diffusion models gradually introduce noise through iterations, aiming to learn the information decay caused by noise and generate images based on the learned patterns. Diffusion models have achieved outstanding performance in high-resolution image generation [35], surpassing GANs in terms of image synthesis quality and diversity [36]. The success of diffusion models has attracted significant attention from researchers, leading to further improvements in generation quality [37], [38] and sampling speed [39], [40], [41].

Recently, conditional diffusion models [36], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54] have been developed and applied to various downstream tasks. Numerous diffusion-based methods for image-to-image translation [43], [51], [52], [53], [54] have been proposed, targeting low-level visual tasks such as colorization, denoising, inpainting, and super-resolution. Cross-modal works have also demonstrated the remarkable performance of diffusion models in text-to-image generation [46], [47], [48], [49], [50]. Furthermore, diffusion models have shown remarkable results in semantic image synthesis [44], image segmentation [55], video generation [56], and pose estimation [45], among other specific tasks. There is very little work on applying diffusion models to cloud removal, and the representative one we could find is DDPM-CR [29]. However, DDPM-CR directly applies DDPM [35] to cloud removal tasks without any adaptation. For example, DDPM-CR mixes noisy images and conditional images for encoding, uses the original UNet model from [36], and uses noise-based regression methods, which severely limit the performance and efficiency of cloud removal. Inspired by these works, we pioneer the exploration of using diffusion models to tackle the cloud removal task, proposing various improvement strategies to address the complexities of cloud removal and achieve high-fidelity cloud-free images.

III. METHOD

A. Problem Definition

Cloud removal aims to generate a cloud-free image from cloudy optical satellite images. We define the cloudy images as $\mathbf{x} \in \mathbb{R}^{N\times C\times H\times W}$, where C, H, and W represent the number of channels, height, and width of the satellite image, respectively, and N represents the number of images. Specifically, when $N = 1$, indicating that only one cloudy image is input, we refer to it as single-temporal cloud removal; when $N > 1$, the input consists of multiple cloudy images acquired at different times but at the same geographical location, and we refer to it as multitemporal cloud removal. The cloud-free images corresponding to each area are denoted as $\mathbf{y} \in \mathbb{R}^{1\times C\times H\times W}$.

B. Overall Framework

In this article, we propose a novel framework, called DiffCR, combined conditional diffusion models [35] and deep convolutional network for cloud removal. As shown in Fig. 1, it consists of two parts: the noise injection process from left to right and the denoising process from right to left. Given the multiple cloudy images $\mathbf{x} \in \mathbb{R}^{N\times C\times H\times W}$ as a condition and the degraded image $\mathbf{y} \in \mathbb{R}^{1\times C\times H\times W}$ with a certain noise level t added to the ground truth image $\mathbf{y}_0$, our framework can iteratively refine and generate high-quality cloud-free images with fine-grained details. DiffCR is inspired by current diffusion models. The purpose is to simulate the diffusion process in thermodynamics, to gradually achieve stable performance improvement of cloud removal in the model, rather than directly falling into a local optimum.

1) Forward Diffusion: Forward diffusion is a deterministic process that does not require learning. It follows the Markov assumption and progressively adds noise to the cloud-free image $\mathbf{y}_0$ over T iterations:

$$q(\mathbf{y}_{1:T}|\mathbf{y}_0) = \prod_{t=1}^{T} q(\mathbf{y}_t|\mathbf{y}_{t-1}) \tag{1}$$

$$q(\mathbf{y}_t|\mathbf{y}_{t-1}) = \mathcal{N}\left(\mathbf{y}_t; \sqrt{1-\beta_t}\mathbf{y}_{t-1}, \beta_t\mathbf{I}\right) \tag{2}$$

where β_t is a predefined hyperparameter representing the noise variance. Typically, β_t ranges from 0 to 1 and satisfies $\beta_1 < \beta_2 < \cdots < \beta_T$. As T increases, $\mathbf{y}_T$ gradually loses its original

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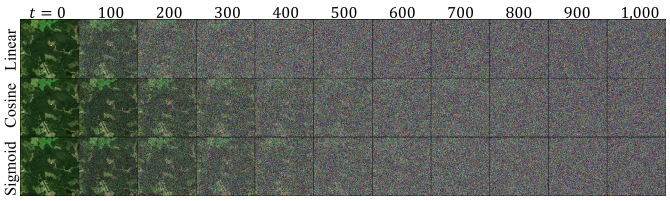


Fig. 2. Samples generated by the diffusion process with different noise schedules. The linear [35] schedule produces samples that are almost pure noise in the end, with little difference. In contrast, the cosine [57] and sigmoid [58] schedules slowly add noise and generate high-quality samples, which are more beneficial for training the diffusion-based denoising model.

data characteristics. When $T \to \infty$, $\mathbf{y}_T$ becomes a random noise that follows a Gaussian distribution. We refer to the variances at different iterations as the noise schedule.

In fact, through mathematical derivation, we can directly sample $\mathbf{y}_t$ at any step t based on the original data $\mathbf{y}_0$

$$q(\mathbf{y}_t|\mathbf{y}_0) = \mathcal{N}\left(\mathbf{y}_t; \sqrt{\overline{\alpha}_t}\mathbf{y}_0, (1-\overline{\alpha}_t)\mathbf{I}\right) \tag{3}$$

where $\alpha_t = 1 - \beta_t$ and $\overline{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. This is a crucial property, and from (3), we can see that $\mathbf{y}_t$ is actually a linear combination of $\mathbf{y}_0$ and Gaussian noise. Furthermore, we can define the noise schedule based on $\overline{\alpha}_t$ instead of β_t, such as [57] and [58]. An effective noise schedule can make the diffusion process more natural, accelerate model convergence, and improve performance. Fig. 2 illustrates the visual effects of different noise schedule strategies in the diffusion process. In Section IV-C1, we validate the impact of different noise schedules on cloud removal performance.

2) Reverse Denoising: If we can reverse the diffusion process, i.e., the sample from $q(\mathbf{y}_{t-1}|\mathbf{y}_t)$, we can reconstruct an accurate cloud-free sample from a random Gaussian distribution $\mathcal{N}(0, \mathbf{I})$. However, since it is not feasible to directly predict $q(\mathbf{y}_{t-1}|\mathbf{y}_t)$ from the complete dataset, we generally use a parameterized neural network model p_θ to approximate the conditional probability distribution and perform the reverse denoising process (Algorithm 1)

$$p_\theta(\mathbf{y}_{0:T}) = p(\mathbf{y}_T) \prod_{t=1}^{T} p_\theta(\mathbf{y}_{t-1}|\mathbf{y}_t, \mathbf{x}) \tag{4}$$

$$p_\theta(\mathbf{y}_{t-1}|\mathbf{y}_t, \mathbf{x}) = \mathcal{N}\left(\mathbf{y}_{t-1}; \boldsymbol{\mu}_{\theta(\mathbf{y}_t,t,\mathbf{x}),\boldsymbol{\Sigma}_\theta(\mathbf{y}_t,t,\mathbf{x})}\right) \tag{5}$$

where $\mathbf{y}_t$, t, and $\mathbf{x}$ are inputs to the model and $\boldsymbol{\mu}_\theta$ and $\boldsymbol{\Sigma}_\theta$ represent the mean and variance of the prior distribution predicted by the model, respectively.

Algorithm 1 Training a Conditional Denoising Model

Input: Cloudy images $\mathbf{x}$, noisy cloud-free image $\mathbf{y}_t$, noise level t

Output: $\hat{\mathbf{y}}_0$ estimated by conditional denoising model $\boldsymbol{f}_\theta$

repeat
  $(\mathbf{x}, \mathbf{y}_0) \sim q(\mathbf{x}, \mathbf{y})$
  $t \sim \text{Uniform}(\{1, \ldots, T\})$
  $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  Take a gradient descent step on
  $\nabla_\theta \left\| \mathbf{y}_0 - \boldsymbol{f}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{y}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t, \mathbf{x}) \right\|_p^p$
until converged

It is noteworthy that inference (Algorithm 2) is also a reverse Markov process, but it starts from Gaussian noise and samples forward step by step, i.e., it generates high-fidelity cloud-free images through iterative refinement. Theoretically, the larger the time step T for adding noise, the higher the quality of the final inferred image generated. Existing diffusion models [44], [52], [53] typically adopt the sampling schemes proposed in [35] or [39], often requiring thousands of steps to generate high-quality images, which severely affects the practical application of the model, especially on edge devices with low resources. To address this challenge, we adopt a fast ODE solver [41][1] that can generate high-quality images in about ten steps. In the ablation study (Section IV-C4), we provide quantitative and qualitative comparisons of the results from different sampling steps of various samplers.

Algorithm 2 Inference With Iterative Refinement

Input: Cloudy images $\mathbf{x}$, Gaussian noise $\mathbf{y}_T$, noise level t

Output: $\mathbf{y}_0$ estimated by pre-trained conditional denoising model

$\mathbf{y}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, \ldots, 1$ **do**
  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
  $\mathbf{y}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{y}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{f}_\theta(\mathbf{y}_t, t, \mathbf{x})\right) + \sqrt{1-\alpha_t}\mathbf{z}$
end for
return $\mathbf{y}_0$

C. Conditional Denoising Model

From the perspective of decoupling feature extraction, we present a novel conditional denoising model based on our carefully designed TCFBlock as the fundamental component, as illustrated in Fig. 3. Decoupled feature extraction can decompose the information in the original feature space into multiple independent factors, which can better describe different aspects of the data, thereby improving the model's generalization ability and interpretability. The model consists of three parts: 1) a condition encoder; 2) a time encoder; and 3) a denoising autoencoder. The condition encoder and the time encoder are responsible for extracting the multiscale spatial features of cloudy images $\mathbf{x}$ and temporal features of noise level t, respectively, which are then fed into the denoising autoencoder to guide the entire reconstruction process. Under the supervision of the restoration loss, the model outputs high-fidelity cloud-free images.

1) Time and Condition Fusion Block: It is well known that the training of diffusion models typically involves many iterative steps. These require substantial computational resources. Motivated by this and inspired by the network structures in [36], [59], [60], and [61], we design an efficient TCFBlock as the basic unit of the conditional denoising model. It significantly reduces computational complexity while achieving the SOTA performance. As illustrated in Fig. 4, TCFBlock comprises the spatial extraction module (SEM) and the feature

[1]Please refer to the supplementary material for code implementation details.

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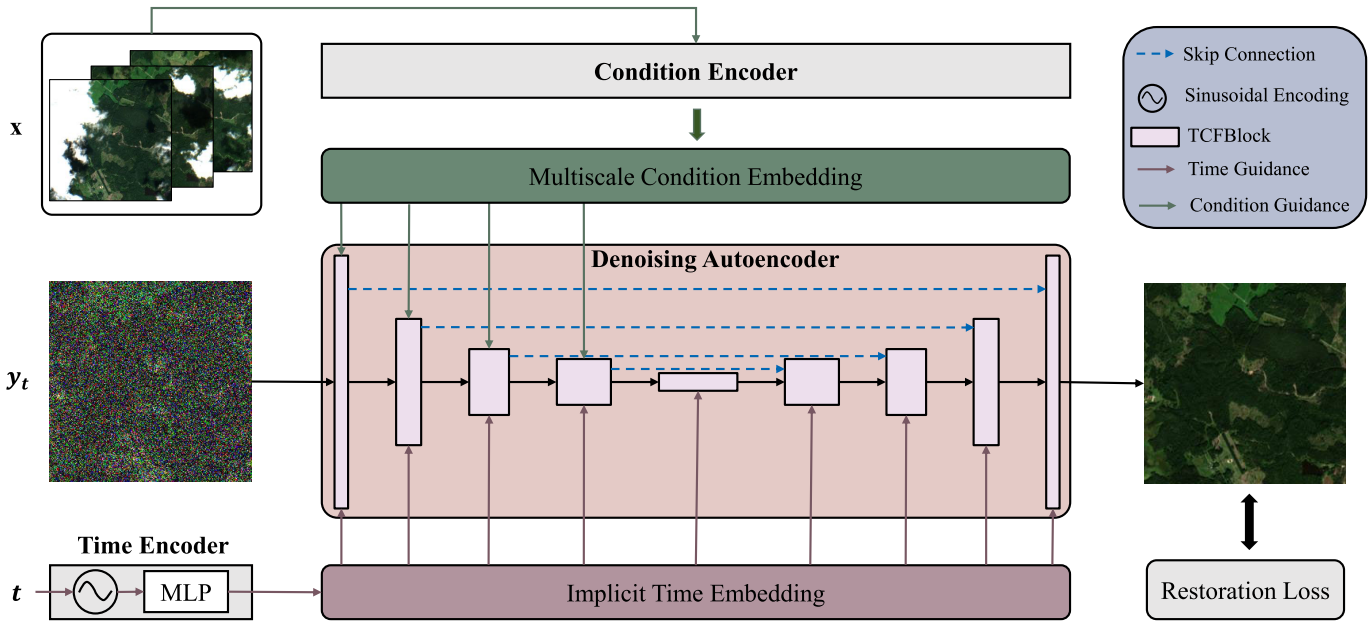


Fig. 3. Overall architecture of the conditional denoising model of DiffCR. The model consists of three parts: 1) a condition encoder; 2) a time encoder; and 3) a denoising autoencoder. The condition encoder and the time encoder are responsible for extracting the spatial features of cloudy images $\mathbf{x}$ and temporal features of noise level t, respectively, which are then fed into the denoising autoencoder to guide the entire denoising process. The condition encoder and denoising autoencoder comprise many of our developed TCFBlocks (refer to Fig. 4). The time encoder comprises a sinusoidal encoding function and an MLP. Ultimately, the data distribution of cloud-free images $\mathbf{y_0}$ is estimated under the supervision of the restoration loss.

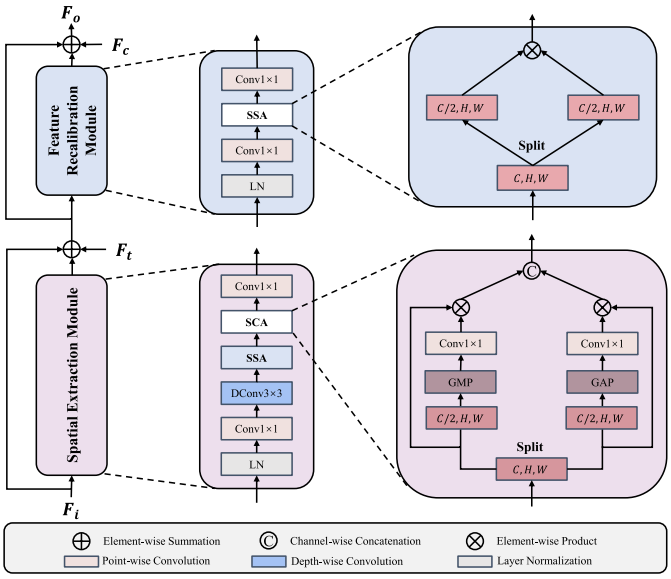


Fig. 4. Schematic of our proposed TCFBlock.

recalibration module (FRM)

$$\mathbf{F}_n = \mathrm{SEM}(\mathbf{F}_i) + \mathbf{F}_i + \mathbf{F}_t \tag{6}$$

$$\mathbf{F}_o = \mathrm{FRM}(\mathbf{F}_n) + \mathbf{F}_n + \mathbf{F}_c \tag{7}$$

where $\mathbf{F}_i \in \mathbb{R}^{C\times H\times W}$ represents the input features, $\mathbf{F}_n \in \mathbb{R}^{C\times H\times W}$ represents the output of the feature by SEM, $\mathbf{F}_c \in \mathbb{R}^{C\times H\times W}$ represents the spatial features of cloudy images, $\mathbf{F}_t \in \mathbb{R}^{C\times 1\times 1}$ represents the temporal features of noise level, and $\mathbf{F}_o \in \mathbb{R}^{C\times H\times W}$ is the final output features through FRM.

a) Spatial extraction module: SEM is designed to extract fine-grained spatial features and is constructed as an inverted bottleneck structure. First, layer normalization [62] commonly used in Transformers [63], [64], [65] is employed to replace batch normalization [66] for normalizing input features, aiming to stabilize training. Then, a 1×1 convolution with channel expansion and a depthwise separable convolution with a 3×3 kernel are utilized to extract detailed spatial features. Subsequently, a split self-activation (SSA) mechanism [60] is introduced to replace the role of conventional activation functions, thereby enhancing the nonlinear expressive power of the model

$$\mathbf{F}_g = \mathrm{SSA}(\mathrm{DConv}_{3\times 3}(\mathrm{Conv}_{1\times 1}(\phi(\mathbf{F}_i)))) \tag{8}$$

where ϕ represents normalization and Conv and DConv, respectively, denote standard and depthwise separable convolution.

In recent years, channel attention mechanisms [67], [68], [69] have achieved great success in various computer vision tasks. Therefore, we introduce a novel and efficient split channel attention (SCA) into SEM to enhance the robustness of the extracted features. As shown in Fig. 4, SCA comprises two channel attention branches: the global channel attention branch and the local channel attention branch. First, the input features are divided into two parts in the channel dimension, followed by global average pooling (GAP) and global maximum pooling (GMP) in each part. Subsequently, a 1×1 convolution is employed to establish explicit correlations between the channels in each part. Finally, the selected features from the two branches are concatenated along the channel dimension to generate robust features that simultaneously consider global and local information. As average pooling emphasizes the extraction of mean information of features and can retain some intricate details of the image, we refer to the branch where it is located as the global channel attention branch. Conversely, since maximum pooling focuses more on extracting the most salient features of the image, we denote

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the branch where it is located as the local channel attention branch. Finally, a 1×1 convolution is introduced to modulate the concatenated features further

$$\mathbf{F}_s = \mathbf{W}_0\big(\mathrm{SCA}\big(\mathbf{F}_g\big)\big) \tag{9}$$

$$\mathrm{SCA}\big(\mathbf{F}_g\big) = \Phi\big(\mathbf{W}_1\big(\mathrm{GAP}\big(\mathbf{F}_{g1}\big)\big), \mathbf{W}_2\big(\mathrm{GMP}\big(\mathbf{F}_{g2}\big)\big)\big) \tag{10}$$

where $\mathbf{F}_{g1} \in \mathbb{R}^{(C/2)\times H\times W}$ and $\mathbf{F}_{g2} \in \mathbb{R}^{(C/2)\times H\times W}$ are the results of splitting $\mathbf{F}_g \in \mathbb{R}^{C\times H\times W}$ in the channel dimension. $\mathbf{W}_0 \in \mathbb{R}^{C\times C}$, $\mathbf{W}_1 \in \mathbb{R}^{(C/2)\times(C/2)}$, and $\mathbf{W}_2 \in \mathbb{R}^{(C/2)\times(C/2)}$ denote the weight matrices of the pointwise convolution. Φ represents the channel-wise concatenation operation.

b) Feature recalibration module: FRM is designed to recalibrate the spatial features of the conditional and noisy images, the temporal features of the noise level, which consists of two pointwise convolution layers, one activation layer, and one normalization layer [see (11)]. To reduce computational complexity, we adopt SSA [60] to replace the commonly used ReLU [70] or SiLU [71] activation functions in deep models. In [61], it was found that fewer activation functions can lead to better performance. Therefore, we employ only a single SSA to enhance the model's nonlinearity. In addition, the inverted bottleneck structure [72], which has larger middle and smaller ends, has been proven to prevent information loss effectively. Hence, we use it in the two pointwise convolutions instead of the bottleneck structure [59]

$$\mathbf{F}_r = \mathbf{V}_2(\sigma(\mathbf{V}_1(\phi(\mathbf{F}_m))) \tag{11}$$

where $\mathbf{F}_r \in \mathbb{R}^{C\times H\times W}$ represents the FRM recalibrated features, σ denotes the activation function, and ϕ refers to layer normalization. $\mathbf{V}_1 \in \mathbb{R}^{C\times 2C}$ and $\mathbf{V}_2 \in \mathbb{R}^{2C\times C}$, respectively, denote the weight matrices of the two pointwise convolutions in the inverted bottleneck structures.

2) Condition Encoder: Existing conditional diffusion models [52], [53] directly concatenate the conditional input and noisy input along the channel dimension and feed them into the denoising model, which increases the difficulty of model training. Motivated by this, we develop a lightweight decoupled CNN-based condition encoder, which is composed of several TCFBlock (without $\mathbf{F}_t$ and $\mathbf{F}_c$ injection) followed by downsampling layers to explicitly extract multiscale features of the conditional image $\mathbf{x}$. Specifically, for the conditional input $\mathbf{x}$, we first utilize the TCFBlock to refine the spatial relationship and then perform scale reduction. Following [65], we introduce convolutional layers with kernel and stride size of 2×2 for downsampling, and to mitigate inherent information loss during downsampling, we increase the number of channels

$$\mathbf{F}_c = \mathrm{DownSample}(\mathrm{TCFBlock}(\mathbf{x})) \tag{12}$$

where DownSample$(\cdot)$ refers to the convolution mentioned above and $\mathbf{x}$ and $\mathbf{F}_c$ denote the input conditional image and the output feature after scale reduction, respectively.

3) Time Encoder: To prevent the denoising autoencoder from learning a fixed paradigm, we choose to add images with varying levels of random noise during input. Correspondingly, we need to include the time step t that reflects the level of noise as an additional input for the denoising autoencoder to guide the entire learning process. Specifically, for the time step t, we adopt the same time encoder (13) as in [35], [36], [52], and [53]. First, we perform a sinusoidal encoding and feed it to a multilayer perceptron to obtain an implicit time embedding. Finally, after broadcasting its dimensions, the time encoding is injected into each TCFBlock of the denoising autoencoder. Since the time step t does not contain spatial information, convolution operations similar to those used for images are unnecessary. These operations fail to bring additional performance gains and increase computational overhead

$$\mathbf{F}_t = \mathrm{Reshape}(\mathrm{MLP}(\sin(t))) \tag{13}$$

where t represents the time step, $\sin(\cdot)$ denotes the sinusoidal encoding, and MLP represents a multilayer perceptron (MLP) consisting of two linear layers with reverse bottleneck structure and a SiLU [71] activation function. Reshape$(\cdot)$ denotes broadcasting the time embedding to modulate weights of image features, and $\mathbf{F}_t$ represents implicit time embedding.

4) Denoising Autoencoder: We utilize the TCFBlock in Section III-C1 as the basic unit to design a UNet-structured [73] autoencoder for the denoising process of DiffCR.

a) Encoder: The encoder comprises an array of consecutively stacked TCFBlocks and several downsampling layers. This design aims to extract diverse multiscale features at varying resolutions, encapsulated by (14). Informed in [65], we adopt a downsampling scheme using convolutional layers characterized by a 2×2 kernel size and stride. To mitigate the inherent information loss during the downsampling operation, we simultaneously expand the number of channels, which essentially enhances the feature representation capacity of each layer

$$\mathbf{F}_{e+1} = \mathrm{DownSample}(\mathrm{TCFBlock}(\mathbf{F}_e, \mathbf{F}_c\mathbf{F}_t)) \tag{14}$$

where $\mathbf{F}_e \in \mathbb{R}^{C\times H/e\times W/e}$ and $\mathbf{F}_{e+1} \in \mathbb{R}^{C\times H/(e+1)\times W/(e+1)}$, respectively, denote the feature maps extracted from the eth and $(e+1)$th layers. DownSample$(\cdot)$ operation embodies the convolutional downsampling approach, further enhancing the hierarchical representation of our model. By progressively accumulating abstract features layer by layer, our encoder provides a robust and rich feature set that boosts the performance of downstream tasks.

b) Middle: The purpose of the middle block is to modulate the features extracted by the encoder at the minimum spatial scale, further refining the features under the guidance of a large receptive field or establishing spatial positional correlations among the features. Therefore, we can inherit the encoder's design by stacking multiple TCFBlocks continuously

$$\mathbf{F}_{m+1} = \mathrm{TCFBlock}(\mathbf{F}_m, \mathbf{F}_c, \mathbf{F}_t). \tag{15}$$

c) Decoder: The design of the decoder is similar to the encoder, where multiple TCFBlocks are stacked, and upsampling layers are utilized to gradually restore the latent features to the spatial scale of the original input image. To mitigate information loss, we leverage PixelShuffle [74] as a replacement for interpolation or transposed convolution for

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upsampling. Similar to UNet [73], we also introduce skip connections. However, instead of using the standard channel-wise concatenation operation, we directly use element-wise addition. The skip connections with direct addition do not introduce additional parameters, whereas channel-wise concatenation requires learning additional parameters to adjust the dimensionality of the feature channels. When the low-level features are directly added to the high-level features, the low-level features can be more directly fused with the high-level features, while channel-wise concatenation may require more complex transformations to achieve feature fusion

$$\mathbf{F}_{d+1} = \text{TCFBlock}(\text{UpSample}(\mathbf{F}_d) + \mathbf{F}_e, \mathbf{F}_c, \mathbf{F}_t). \tag{16}$$

D. Regression Target and Loss Function

The existing unconditional diffusion models [35] and conditional diffusion models [36], [42], [44], [52], [53] commonly adopt the noise prediction approach by regressing the output of f_θ to ϵ, wherein the neural network model is trained to learn the Gaussian noise added at each step of the forward process

$$\mathcal{L}_{t-1}^{\text{simple}} = \mathbb{E}_{\mathbf{y}_0,\epsilon}\left\|\epsilon - f_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{y}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t, \mathbf{x}\right)\right\|_p^p \tag{17}$$

where f_θ represents the parameterized denoising model, $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ represents randomly sampled noise from a Gaussian distribution, and $\mathbf{x}$ and $\mathbf{y}$ represent samples from the dataset.

However, we observe that the noise prediction method increases training difficulty and can even lead to mode collapse when there is abundant conditional input information. Motivated by this consideration, we propose a novel and more straightforward prediction approach by regressing the output of f_θ to $\mathbf{y}_0$, termed "data prediction," as a replacement for noise prediction

$$\mathcal{L}_{t-1}^{\text{simple}} = \mathbb{E}_{\mathbf{y}_0,\epsilon}\left\|\mathbf{y}_0 - f_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{y}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t, \mathbf{x}\right)\right\|_p^p. \tag{18}$$

Although the values of ϵ and $\mathbf{y}_0$ can be determined by each other, changing the regression target affects the scale of the loss function, thereby influencing model training. Since the conditional input image and the cloud-free ground truth image share more similar data distributions, data prediction is significantly easier than noise prediction, resulting in improved model convergence and cloud removal performance. This point is further validated through ablation experiments in Section IV-C2. We compare these two different regression methods and find that directly predicting $\mathbf{y}_0$ (our proposed data prediction) leads to easier convergence, significantly when the number of condition $\mathbf{x}$ is increased. It is worth noting that DALL.E 2 [49] developed by OpenAI also reports a similar finding in their work.

IV. Experiments

A. Datasets

We perform experiments on three large-scale cloud removal datasets for optical satellite images: Sen2_MTC_New [15], WHUS2-CRv [17], and SEN12MS-CR [18].

1) Sen2_MTC_New: To address the issues of poor annotation quality and low sample resolution in the dataset of [14], Huang et al. [15] constructed a new Sentinel-2 satellite image dataset for training multitemporal cloud removal models. This dataset contains approximately 50 nonoverlapping tiles, and each tile consists of around 70 pairs of cropped 256 × 256 pixel patches. Similar to the dataset of [14], Sen2_MTC_New follows the same procedure to construct multitemporal image pairs, where three cloudy images at the same location correspond to one cloud-free image. Sen2_MTC_New preserved the images in their original TIFF format, with pixel value ranging from 0 to 10 000.

2) WHUS2-CRv: This dataset serves as a comprehensive and representative resource for thin cloud removal, boasting features such as frequent data acquisitions, global coverage, extensive land area representation, and complete seasonal diversity. The dataset comprises pairs of cloud and cloud-free images meticulously chosen with a ten-day interval to minimize reflectance variations between the two. All images are cropped into nonoverlapping patches. Given the varying spatial resolutions of bands (10 m for VNIR, 20 m for VRE/NIRn/SWIR, and 60 m for Ca/Wv/Cir), patches were standardized to sizes of 384 × 384, 192 × 192, and 64 × 64 for 10-, 20-, and 60-m bands, respectively. Each group of patches is accompanied by three multispectral images, ensuring a coherent representation.

3) SEN12MS-CR: The dataset is the first public dataset for cloud removal in Earth observation and consists of a triplet, featuring orthorectified, georeferenced representations of cloudy, cloud-free, and 13-band multispectral Sentinel-2 images, accompanied by their correspondent Sentinel-1 image. The collected samples are distributed worldwide and in different seasons and contain 169 regions of interest (ROIs). Each scene in the dataset is divided into 256 × 256 pixels patches with 50% spatial overlap between adjacent patches. More than 700 patches were generated for each ROI. The acquisition of paired images across the three modalities is meticulously synchronized within the same meteorological season to mitigate potential surface changes. The training split of the dataset comprises a substantial 114 325 patches, uniformly sampled across all continents and seasons. The test split consists of 7893 geospatially separated images, strategically sampled from ten distinct ROIs spanning across diverse continents and meteorological seasons.

B. Implementation Details

To facilitate experimental reproducibility, this section presents a detailed account of the hyperparameter configurations employed during the training process and the metrics utilized to evaluate the experimental outcomes. All the experiments were conducted on a workstation equipped with two AMD EPYC 7402 24-core processors, 128 GB of memory, and four NVIDIA GeForce RTX 3090 GPUs with 24 GB of memory. The operating system used was Ubuntu 22.04, and all networks were implemented in PyTorch 1.10.1 with CUDA 11.1.

1) Sen2_MTC_New: We selected AdamW [75] as our optimizer with an initial learning rate of $5e^{-5}$ and a weight decay

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TABLE II
QUANTITATIVE COMPARISON OF CLOUD REMOVAL PERFORMANCE AND EFFICIENCY BETWEEN DIFFCR AND EXISTING MODELS ON THE SEN_MTC_NEW DATASET. N REPRESENTS THE NUMBER OF INPUT CLOUDY IMAGES

Method	$N=2$				$N=3$				Params (M) ↓	MACs (G) ↓
	PSNR ↑	SSIM ↑	FID ↓	LPIPS ↓	PSNR ↑	SSIM ↑	FID ↓	LPIPS ↓		
MCGAN [12]CVPRW17	14.567	0.314	214.057	0.647	17.448	0.513	147.057	0.447	4.42	71.56
Pix2Pix [13]CVPR17	14.199	0.256	233.524	0.736	16.985	0.455	164.524	0.535	11.41	58.94
AE [24]ECTI-CON18	12.250	0.245	275.134	0.803	15.100	0.441	206.134	0.602	6.53	**35.72**
STNet [25]TGRS20	13.651	0.228	231.683	0.704	16.206	0.427	161.683	0.503	4.64	304.31
DSen2-CR [27]ISPRS J PHOTOGRAM20	13.989	0.337	208.208	0.647	16.827	0.534	140.208	0.446	18.92	1240.23
STGAN [14]WACV20	15.247	0.392	250.150	0.714	18.152	0.587	182.150	0.513	231.93	1094.94
CTGAN [15]ICIP22	15.807	0.416	198.704	0.584	18.308	0.609	128.704	0.384	642.92	632.05
CR-TS-Net [16]TGRS22	15.953	0.425	163.364	0.542	18.585	0.615	96.364	0.342	38.68	7602.97
PMAA [26]ECAI23	15.624	0.415	184.214	0.593	18.369	0.614	118.214	0.392	3.45	92.34
UnCRtainTS [28]CVPRW23	15.783	0.432	161.509	0.533	18.770	0.631	93.509	0.333	0.56	37.16
DDPM-CR [29]REMOTE SENS-BASEL23	15.846	0.420	161.401	0.530	18.742	0.614	94.401	0.329	445.44	852.37
DiffCR [Ours]	**16.222**	**0.444**	**150.771**	**0.484**	**19.150**	**0.671**	**83.162**	**0.291**	22.91	45.86

of 0. During the training phase, we employed batch sizes of 8. The training process was conducted over a total of about 7M steps. The weight initialization in this work employs the Kaiming normal distribution [76]. An exponential moving average (EMA) [77] is applied to the model parameters, starting from the first iteration of training and updated after each training iteration with a decay rate of 0.9999, which helps stabilize the training process and prevent overfitting. During diffusion progress, we set the maximum noise step to 2000. For MCGAN [12] and UnCRtainTS [28] that are not specifically designed for multitemporal cloud removal, we modified their channels of the input and output layers. For multitemporal inputs, we stack them in the channel dimension and feed them to these methods to achieve a fair comparison. For the DSen2-CR [27], a model developer for SAR and optical inputs, we removed SAR input to match the dataset with only Sentinel-2 images. In addition, we also added the fair comparison of DiffCR and other baseline methods on two images ($N = 2$) in Table II for the usual case where only two images are available.

2) WHUS2-CRv: We followed the training parameters of CR4S2 [17] to train the proposed DiffCR for a fair comparison. Adam [78] optimizer is adopted using $\beta_1 = 0.9$, $\beta_2 = 0.999$, the initial learning rate is set to $2e^{-4}$, and the exponential decay at decay rate is 0.96. We adjusted the input and output layers of DiffCR to match the input and output of the multispectral bands.

3) SEN12MS-CR: We reproduced our experimental results based on the open-source code of UnCRtainTS [28]. The input contains a 13-channel multispectral image and a 2-channel SAR image, and the output contains a 13-channel multispectral image, so we modify the channel number of the input and output layers of DiffCR to match. For a fair comparison, we also refer to the training configuration of UnCRtainTS [28]. The number of training epochs is 20, the batch size is 4, and an Adam [78] optimizer with an initial learning rate of $1e^{-3}$.

For the Sen2_MTC_New dataset, We report the peak signal-to-noise ratio (PSNR), dB, structural similarity index measure (SSIM) [79], learned perceptual image patch similarity (LPIPS) [80], and Frechet inception distance (FID) [81] of the test set to evaluate the quality of the generated cloud-free images. It is noteworthy that PSNR and SSIM measure the differences between images on a pixel basis, while FID and LPIPS measure the differences between images based on deep feature vectors. Compared to traditional image quality assessment metrics (PSNR and SSIM), FID and LPIPS align more with human visual perception. Moreover, LPIPS can be almost considered a regional FID metric, as even when FID varies greatly, LPIPS scores show minimal changes. Therefore, when the LPIPS index of two methods is almost the same, the superiority of the method can be judged by comparing the FID. For the evaluation of model efficiency, we report the number of parameters (Params, M) and multiply–accumulate operations (MACs, G) for all models, which are calculated via the open-source tool.[2] For the WHUS2-CRv dataset, we followed the previous SOTA model (CR4S2 [17]) to report the results of PSNR and SSIM. For the SEN12MS-CR dataset, we followed the previous SOTA model (UnCRtainTS) to report the results of mean absolute error (MAE), spectral angle mapper (SAM) [82], PSNR, and SSIM.

C. Ablation Studies

In this section, we present extensive ablation studies to validate the effectiveness of the proposed DiffCR and its constituent parts. Specifically, we investigate the impact of the following components: 1) the noise schedule in the forward diffusion; 2) the regression target of the denoising model; 3) the processing method for conditional cloudy images; 4) sampling steps during the inference phase; and 5) changes in the architecture of the denoising model. All ablation experiments are conducted on the Sen2_MTC_New dataset.

1) Does Noise Schedule Matter: The results in Table III present the impact of different noise schedules during the diffusion process. It can be observed that the sigmoid schedule achieves the best cloud removal performance, with a slight decrease in performance for the cosine schedule, and the linear schedule exhibits the poorest performance, which is consistent with theoretical expectations described in Section III-B1.

[2]https://github.com/Lyken17/pytorch-OpCounter

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TABLE III

QUANTITATIVE COMPARISON OF THE CLOUD REMOVAL PERFORMANCE OF NOISE SCHEDULES SELECTED DURING THE DIFFUSION PROCESS OF DIFFCR

Schedule	PSNR ↑	SSIM ↑	FID ↓	LPIPS ↓
Linear [35]	19.060	0.663	92.167	0.294
Cosine [57]	19.148	0.665	84.940	0.293
Sigmoid [58]	**19.150**	**0.671**	**83.162**	**0.291**

TABLE IV

QUANTITATIVE COMPARISON OF THE CLOUD REMOVAL PERFORMANCE BETWEEN NOISE PREDICTION AND DATA PREDICTION DURING THE REVERSE PROCESS OF DIFFCR. N REPRESENTS THE NUMBER OF CONDITIONAL CLOUDY IMAGES

Regression	N	PSNR ↑	SSIM ↑	FID ↓	LPIPS ↓
Noise Prediction [35]	1	3.738	0.215	410.835	0.924
	2	3.711	0.106	504.806	1.145
	3	3.641	0.099	511.454	1.205
Data Prediction [Ours]	1	15.139	0.433	154.243	0.547
	2	16.222	0.444	150.771	0.484
	3	**19.150**	**0.671**	**83.162**	**0.291**

TABLE V

QUANTITATIVE COMPARISON OF THE CLOUD REMOVAL PERFORMANCE OF DIFFERENT PROCESSING METHODS OF CONDITIONAL CLOUDY IMAGES DURING THE REVERSE PROCESS OF DIFFCR

Condition Process	PSNR ↑	SSIM ↑	FID ↓	LPIPS ↓
Concat [29], [53]	18.821	0.638	98.387	0.342
Decouple [Ours]	**19.150**	**0.671**	**83.162**	**0.291**
Δ	+ 0.329	+ 0.033	- 15.225	- 0.051

TABLE VI

QUANTITATIVE COMPARISON OF THE CLOUD REMOVAL PERFORMANCE AND EFFICIENCY OF DIFFERENT SAMPLING STEPS DURING THE INFERENCE PHASE OF DIFFCR

Step	Latency (s)	PSNR ↑	SSIM ↑	FID ↓	LPIPS ↓
1	0.09	19.150	0.671	83.162	0.291
3	0.16	**19.182**	**0.674**	**82.978**	**0.291**
5	0.23	19.164	0.674	83.281	0.291
10	0.39	19.170	0.672	83.177	0.291
50	1.74	19.102	0.666	84.635	0.292
100	3.39	19.077	0.664	84.998	0.292

Therefore, we recommend using the sigmoid schedule for diffusion-based cloud removal or other generative tasks, as it helps with model convergence and performance improvement.

2) Data Prediction or Noise Prediction: We experimentally verified the impact of the proposed data prediction and native noise prediction on cloud removal performance, as shown in Table IV. As expected (described in Section III-D), data prediction outperforms noise prediction for any value of N in {1, 2, 3}, where N represents the sequence length of cloudy images. In particular, as N increases, data prediction performs increasingly better, while noise prediction performs increasingly worse. This is because, in the data prediction, the model attempts to predict a target (i.e., the cloud-free image after denoising) that is more similar in distribution to the input (the noisy cloud-free image and the cloudy image), especially as the similarity of the distribution increases with increasing N. In contrast, noise typically has an entirely different distribution (e.g., Gaussian noise). Therefore, for a model that attempts to extract useful information from the input, predicting a target that is similar to the input may be more accessible, while noise prediction often leads to mode collapse.

3) Processing of Conditional Cloudy Images: To address the conditional cloudy image processing in the DiffCR, we quantitatively verify two different feature extraction and fusion strategies.

1) *Concat:* The noisy cloud-free image is concatenated with all the cloudy images along the channel dimension and then fed into the denoising model for processing, which is the most straightforward approach adopted by previous works such as [29], [30], [52], and [53].
2) *Decouple:* It is a decoupling condition encoder used in DiffCR.

The experimental results in Table V demonstrate that "Decouple" is the better conditional image processing method, achieving a higher cloud removal performance with improvements of 0.329 in PSNR and 15.225 in FID.

4) Inference Using Different Sampling Steps: The vanilla diffusion model [35] has an unacceptable slow inference speed, requiring thousands of iterations for refining a high-quality image. Even on suitable GPU devices, its inference speed for a single sample can be slower than training for a single epoch, seriously hindering the application of diffusion models in computer vision. The existing diffusion-based cloud removal models [29], [30] do not consider this real-world constriction, requiring many sampling steps during the inference process. In contrast, owing to the design of its framework that leverages data prediction and conditional decoupling, DiffCR can generate high-quality samples in just one step and achieve absolute convergence in 3–5 steps. We compare quantitative results regarding sample quality and inference latency at different sampling steps. The results are summarized in Table VI. It can be observed that the inference latency increases with an increase in the number of sampling steps. Remarkably, excellent results can be obtained with just one sampling step, with marginal performance improvements from 1 to 3 sampling steps and convergence being achieved between three and five steps. However, as the number of sampling steps exceeds 25 steps, the performance of cloud removal, as measured by metrics like LPIPS, gradually declines due to oversampling. Therefore, we ultimately compare the results obtained with one sampling step to other methods, which achieves the SOTA performance with only 0.09 s of inference latency.

5) Architecture Changes in Denoising Model: Since the introduction of the diffusion models [34], [35], most studies have focused on improving their internal diffusion mechanisms to achieve high-quality samples [36], [39] or fast inference speed [41], [56]. However, little research has been done to explore what kind of denoising neural model is more effective in the diffusion framework. Motivated by this, we propose a denoising model baseline with a UNet-like [73] architecture. It consists of several TCFBlocks that we designed. Through comprehensive ablation experiments, we explore the

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TABLE VII
QUANTITATIVE COMPARISON OF CLOUD REMOVAL PERFORMANCE BETWEEN OUR DIFFCR AND EXISTING METHODS ON THE WHUS2-CRV DATASET. "AVG" REPRESENTS THE AVERAGE OF ALL BANDS

Index	Method	B1	B2	B3	B4	B5	B6	B7	B8	B8A	B9	B11	B12	AVG
PSNR	RSC-Net [84]	35.06	34.12	33.51	31.24	32.34	30.51	29.90	29.43	29.32	30.07	29.32	29.40	31.19
	FCTF-Net [85]	37.19	36.57	35.09	33.77	32.81	30.78	30.25	29.85	29.72	30.50	30.69	32.17	32.45
	RSDehazeNet [86]	37.64	37.19	36.08	34.05	33.42	31.48	30.76	29.74	30.20	30.74	30.28	31.50	32.76
	CR4S2 [17]	39.55	38.17	37.05	35.55	34.37	32.15	31.40	31.00	30.96	31.32	31.47	33.31	33.86
	DiffCR [Ours]	**40.91**	**41.78**	**43.37**	**41.58**	**35.27**	**36.68**	**35.66**	**36.26**	**35.82**	**33.53**	**39.42**	**39.60**	**38.32**
SSIM	RSC-Net [84]	0.8459	0.8823	0.9203	0.8919	0.9129	0.8991	0.8930	0.8852	0.8910	0.9010	0.9053	0.8937	0.8935
	FCTF-Net [85]	0.8807	0.9103	0.9286	0.9122	0.9167	0.9031	0.8982	0.8842	0.8959	0.9046	0.9109	0.9105	0.9047
	RSDehazeNet [86]	0.8909	0.9203	0.9387	0.918	0.9221	0.9046	0.9004	0.8893	0.9008	0.9035	0.9145	0.9162	0.9099
	CR4S2 [17]	0.9185	0.9315	0.9443	0.9302	0.9355	0.9236	0.9195	0.9002	0.9200	0.9252	0.9317	0.9334	0.9261
	DiffCR [Ours]	**0.9747**	**0.9888**	**0.9872**	**0.9855**	**0.9665**	**0.9715**	**0.9685**	**0.9816**	**0.9691**	**0.9595**	**0.9625**	**0.9665**	**0.9735**

impact of architectural changes in the denoising model on cloud removal performance, with quantitative results shown in Table I. Specifically, we explore the following architectural changes:

1) the width (channels) of the denoising model;
2) the block types of the denoising model;
3) the depth of stacking multiple blocks;
4) the fusion types of features from skip connections.

As shown in Table I, we find that an increase in depth and width can significantly improve cloud removal performance. Among various block types (such as ResNet [59], ConvNeXt [61], and NAFBlock [60]), our proposed TCFBlock achieves the best performance, and adding more blocks in any part of the encoder, middle layer, or decoder leads to some performance improvement. Notably, the performance improvement is most significant when the encoder and the decoder employ a symmetric structure in terms of depth (as in the third row from the bottom). Moreover, the most straightforward skip connection (Sum, i.e., directly adding encoder and decoder features of the same resolution) achieves the best performance, while more complex skip connections (such as Concat [73], which concatenates features in the channel dimension) lead to training instability, possibly due to the increased learning complexity. Selectively connecting features from the encoder and decoder using attention mechanisms [83] can compromise performance, potentially due to loss of information resulting from such connections, affecting the quality of the generated images. Finally, for a balance between performance and efficiency, we opt for the data represented by the third row of the table as the architecture configuration for the denoising model, as it outperforms all existing methods with lower numbers of parameters and computational complexity.

D. Comparison With SOTA Methods

Tables II, VII, and VIII present the quantitative comparison between our proposed DiffCR and existing models on three datasets Sen2_MTC_New, WHUS2-CRv, and SEN12MS-CR.

1) Performance of Cloud Removal: On three benchmark datasets, DiffCR has achieved consistent SOTA performance on all evaluation metrics of cloud removal, demonstrating the full potential of our proposed method. Specifically, DiffCR achieves an SSIM of 0.444 and 0.671 and an LPIPS of 0.484 and 0.291 on $N = 2$ and $N = 3$ of the Sen2_MTC_New dataset, respectively. In contrast, existing models exhibit much lower performance values. On the WHUS2-CRv dataset, DiffCR achieved an average PSNR of 38.32 and an average SSIM of 0.9735 across the 12 bands. Compared with the previous SOTA method (CR4S2), the PSNR and SSIM achieved 4.46 and 0.0474 improvement, respectively. On the SEN12MS-CR dataset with SAR input, DiffCR obtained an MAE of 0.019, an SAM of 5.821, a PSNR of 31.77, and an SSIM of 0.902. Compared with the previous SOTA method (UnCRtainTS), SAM and PSNR achieved significant improvements of 2.499 and 2.87, respectively. Compared to the existing diffusion-based model DDPM-CR, DiffCR achieves significant performance improvements. Specifically, on the Sen2_MTC_New ($N = 3$) dataset, DiffCR achieves a PSNR of 19.150, which is 0.408 higher than DDPM-CR's PSNR of 18.742, and an LPIPS of 0.291, which is 0.038 lower than DDPM-CR's LPIPS of 0.329. These results demonstrate that DiffCR is superior to the existing diffusion-based model DDPM-CR in terms of accuracy and efficiency.

TABLE VIII
QUANTITATIVE COMPARISON OF CLOUD REMOVAL PERFORMANCE BETWEEN DIFFCR AND EXISTING METHODS ON THE SEN12MS-CR DATASET

Method	MAE ↓	SAM ↓	PSNR ↑	SSIM ↑
McGAN [12]	0.048	15.676	25.14	0.744
SAR-Opt-cGAN [87]	0.043	15.494	25.59	0.764
SAR2OPT [88]	0.042	14.788	25.87	0.793
SpA GAN [21]	0.045	18.085	24.78	0.754
Simulation-Fusion GAN [89]	0.045	16.633	24.73	0.701
DSen2-CR [27]	0.031	9.472	27.76	0.874
GLF-CR [90]	0.028	8.981	28.64	0.885
UnCRtainTS L2 [28]	0.027	8.320	28.90	0.880
DiffCR [Ours]	**0.019**	**5.821**	**31.77**	**0.902**

2) Efficiency of Cloud Removal: It is worth noting that the above superior results are attained while maintaining a reasonable balance between the model's parameters (22.91 M) and the MACs (45.86 G), which is a crucial aspect of the practical application of the model. Compared with models based on GAN or regression, DiffCR surpasses most models in terms of parameter count and computational complexity. Compared with the previous SOTA diffusion-based model DDPM-CR, DiffCR achieves higher performance with only 5.1% and 5.4% of the parameter count and computational complexity, respectively. Therefore, our work allows diffusion

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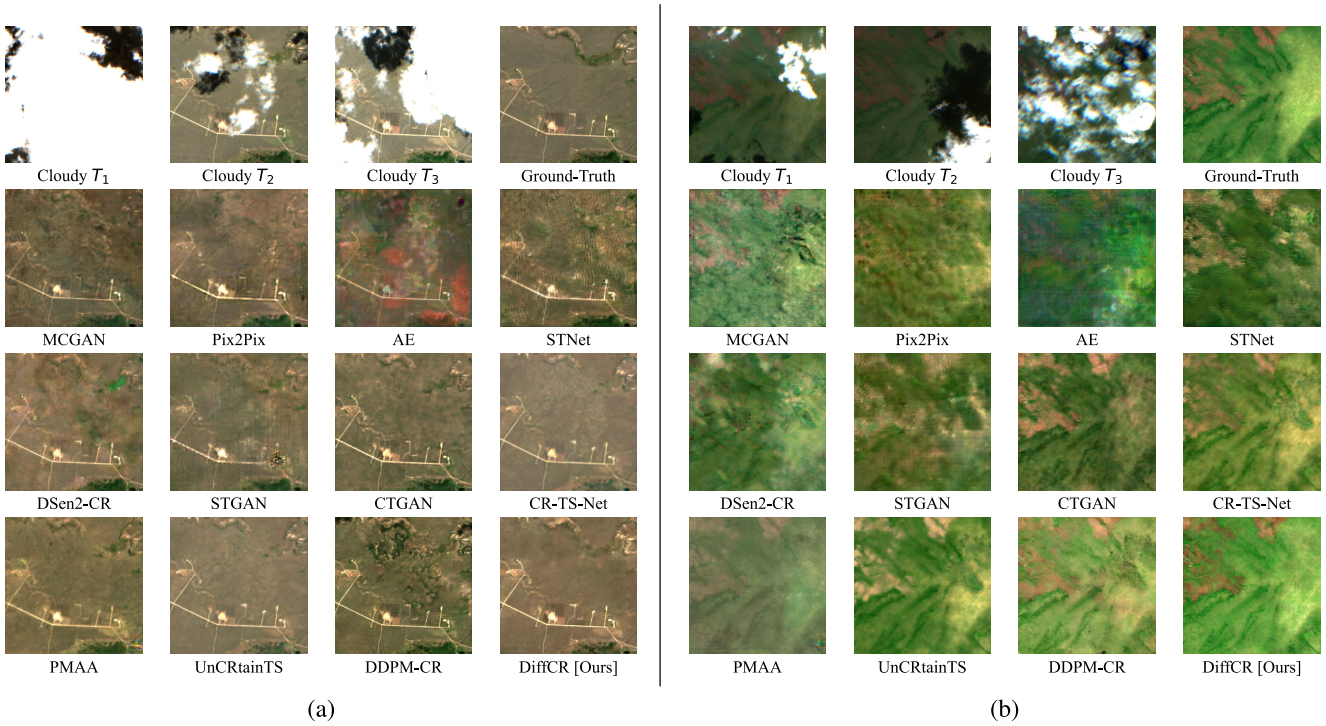


Fig. 5. Qualitative comparison of the visualization results on the Sen2_MTC_New dataset between our proposed DiffCR and other existing methods. (a) Sample of city. (b) Sample of grass.

models to be applied in practical cloud removal tasks for the first time, laying the foundation for future research in this area.

As shown in Fig. 5, we also provide a qualitative comparison between DiffCR and all other existing methods. All samples were randomly selected from the test set. The first row shows the input cloudy image (sequence length of $N = 3$), with the last column representing the reference cloud-free image. The remaining rows display the reconstruction results of different algorithms. It can be observed that DiffCR outperforms all other methods in both local structure generation [as shown in Fig. 5(a)] and global brightness restoration [as shown in Fig. 5(b)], achieving the highest image fidelity.

E. Failure Cases and Limitations

Despite the exemplary performance demonstrated by our method, it still exhibits certain limitations. First, when the input cloudy images containing lake regions with black hole-like shapes, our method encounters challenges in accurately reconstructing the lake areas (see Fig. 6). This is attributed to the inherent similarity between the irregular, dark lake regions and cloud shadows. Second, although our method strives to strike a balance between performance and efficiency by employing lightweight and efficient modules, the inherent mechanisms of the diffusion model necessitate a considerable number of iterative steps during the model training phase in order to achieve convergence. Finally, while our method is designed specifically for the task of cloud removal and has not undergone comprehensive validation across multiple tasks, its inherent generality enables its potential application in diverse domains, such as research pertaining to generative artificial intelligence from text to image and from image to image synthesis.

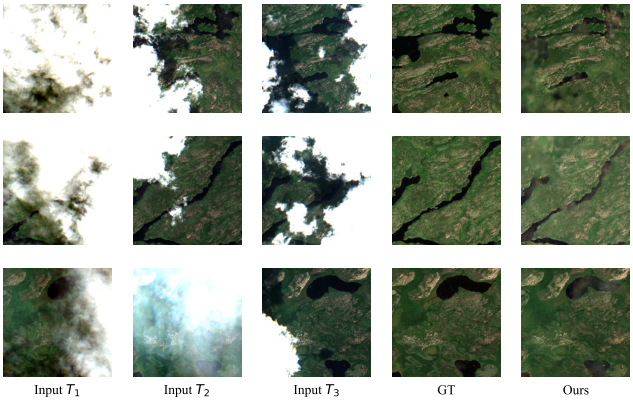


Fig. 6. Three failure examples. DiffCR exhibits inadequacy in scenarios where the input cloudy images feature lake formations resembling black holes. In such cases, the clouds may erroneously occupy the holes in the generation.

V. Conclusion

In this work, we propose DiffCR, a fast conditional diffusion framework for high-quality cloud removal. Existing cloud removal models based on GANs and regression fail to generate high-fidelity images, while diffusion-based models require thousands of iterations to ensure high sampling quality. Thus, balancing sampling speed and sample quality has become the main challenge in cloud removal. In contrast, DiffCR redefines the conditional diffusion framework for cloud removal, designs a novel conditionally decoupled neural model, and parameterizes the denoising model by directly predicting clean data. We also develop a novel and efficient TCFBlock within the model. As a result, DiffCR generates high-fidelity and detail-rich cloud-free images in just one sampling step. Through comprehensive experiments on three

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benchmark datasets, we demonstrate that DiffCR outperforms the best publicly available models regarding synthesis quality.

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Xuechao Zou (Graduate Student Member, IEEE) received the B.E. degree from Qinghai University, Xining, China, in 2021, where he is currently pursuing the M.S. degree with the Department of Computer Technology and Application.

His research interests primarily focus on computer vision, particularly in the area of remote sensing image processing.

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Kai Li (Student Member, IEEE) received the B.E. degree from Qinghai University, Xining, China, in 2020. He is currently pursuing the M.S. degree with the Department of Computer Science and Technology, Tsinghua University, Beijing, China.

His research interests are primarily in audio processing and audiovisual learning, especially speech separation.



Junliang Xing (Senior Member, IEEE) received the dual B.S. degree in computer science and mathematics from Xi'an Jiaotong University, Xi'an, Shaanxi, China, in 2007, and the Ph.D. degree in computer science and technology from Tsinghua University, Beijing, China, in 2012.

He is currently a Professor at the Department of Computer Science and Technology, Tsinghua University. He has published over 100 peer-reviewed articles with more than 12 000 citations from Google Scholar. His research interests mainly focus on computer gaming problems related to single-/multiple-agent learning and human–computer interactive learning.



Yu Zhang received the B.E. degree from Jilin Jianzhu University, Changchun, China, in 2021. He is currently pursuing the M.S. degree with the Department of Computer Technology and Application, Qinghai University, Xining, China.

His research interests include deep single-agent and multiagent reinforcement learning.



Shiying Wang received the B.E. degree from Jilin Agricultural University, Changchun, China, in 2015. She is currently pursuing the Ph.D. degree with Qinghai University, Xining, China.

Her research interests include remote sensing image fusion, especially pan sharpening.



Lei Jin received the bachelor's and Ph.D. degrees from Beijing University of Posts and Telecommunications (BUPT), Beijing, China, in 2015 and 2020, respectively.

He is currently an Associate Research Fellow with BUPT. His research interests include machine learning and pattern recognition, with a focus on 6-DOF pose estimation and human pose estimation. What is more, he concentrated on network security and analyzed traffic security during pursuing his Ph.D. degree.



Pin Tao (Member, IEEE) received the B.S. and Ph.D. degrees in computer science and technology from Tsinghua University, Beijing, China, in 1997 and 2002, respectively.

He is currently an Associate Professor at the Department of Computer Science and Technology, Tsinghua University. He is also the Vice Director of the Key Laboratory of Pervasive Computing, Ministry of Education, Tsinghua University. He has published more than 80 articles and over ten patents. His research interests mainly focus on human–artificial intelligence (AI) hybrid intelligence and multimedia-embedded processing.

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